AI002 – Dataset and Model Research Report

1. Task Details

- Task ID: **AI002**
- Task Name: **Dataset and Model Research**
- Task Lead: **Sam Braley**
- Team Members:
  Trey Christopher Martin
  Thommaya Hewage Vishvani Gunapala
  Sammed Bharat Vaigude
  Mannat
  Sunain Mushtaq
- Sprint: **1**
- Start / End Dates:

  V.1: 21/03/2026 – 27/03/2026
  V.2: 28/03/2026 – 03/04/2026

2. Objective

To identify available datasets and evaluate suitable machine learning models which are pre-trained for the Phoenix project. The goal is to guide early technical decisions and ensure feasibility in later development stages, without the need for extensive training and development by Phoenix. This task should provide the foundation for future Sprints, with clear recommendations on which systems are applicable and which aren't. Ultimately, task AI002 should minimize future interruptions, whilst providing secure dataset and machine learning options that guide all technical decisions.

2.1 Scope

This research focuses on:

- Disaster-related data (e.g., weather, bushfires, floods).
- Cyber threat intelligence data (e.g., phishing, scams).
- Model types suitable for risk scoring and anomaly detection.
- Pre-trained and ready-to-implement Machine Learning models & algorithms.

3. Task Type | Problem Definition

AI002 focuses on data-based research, with a strong emphasis on providing clear and meaningful recommendations on datasets and anomaly detection models. This involves key comparisons, with clear advantages and disadvantages written to suit the development of the Phoenix project. The problem for this task is to identify the most appropriate models that can accurately identify suspicious behaviour and minimize false positives whilst being reliable for the Phoenix project.

The AI system should:

- Detect whether a message or communication is legitimate or a scam.
- Identify anomalous cyber activity during disaster events.
- Estimate cyber risk levels based on disaster context and threat intelligence.

The Machine Learning system should:

- Learn and adapt to new data (flag unusual data)
- Forecast future events based on trends and pattern recognition
- The system should be automated with minimal maintenance and training.
- Classify data

3.1 Problem Types

The primary problem addressed in this project is classification, where messages or communications are categorized as either legitimate or fraudulent. This aligns with the goal of detecting phishing, scams, and misinformation during disaster events.

In addition, anomaly detection is included to identify unusual or suspicious cyber activity that may not be captured by predefined labels. This is particularly important in disaster scenarios where new types of cyber threats may emerge.

Time-series forecasting and probabilistic modelling are not included at this stage, as the current scope focuses on detection rather than prediction or probabilistic risk estimation.

4. Dataset Catalogue

4.1 Overview

To support the identified problem types, the dataset catalogue for this project combines cyber threat intelligence data with disaster-related data. Information on the frequency, location, and intensity of natural hazards can be found in disaster data, such as flood warning datasets, NASA FIRMS fire data, and bushfire incident records (NASA-FIRMS, n.d.; Weather Data Services, 2025). Scam reports, phishing URL datasets, and network intrusion datasets like UNSW-NB15 are examples of cyber data that provide information about malicious activity and unusual behavior in digital systems (GoMask.ai, 2025; Klaas, 2018; Mori Esam, 2025). When combined, these datasets allow for the examination of trends and connections between cyberthreats and disaster occurrences, facilitating efficient risk assessment and forecasting.

4.2 Dataset Table

Dataset Name	Source	Type	Format	Size	Access Method	Quality	Notes
Weather data services	Bureau of Meteorology	Disaster forecast data	RSS. FTP, TXT, PDF, HTML, and XML	Provides forecasts for all states and territories	here Feeds/dataset	High	Required fees for commercial use
FIRMS	NASA	Bushfire forecast data	KML, CSV, text, Shapefiles for archive, RSS, WFS, WMS for RT/URT	Varying between 1k-5k entries for a 7-day forecast archive	here API/dataset	High	Requires an account to access more than the 7-day archive
CFA	CFA	Bushfire forecast data	RSS, XML	Provides updates every ~5 minutes	here Feeds	High	must include links back to CFA when using feeds
VicEmergency	VicEmergency	Disaster forecast data	JSON, XML, RSS	Provides constant updates	here Feeds	High	Must request use of feeds from here
EM-DAT	CRED	Disaster forecast data	XLSX	~27,000 mass disasters	here Archive dataset	High	Required fees for commercial use
Weather API	WeatherAPI	Real-time weather data	JSON, XML	Large (2010–present data)	here API	High	Free for non-commercial use, fast response
Open-Meteo	Open-Meteo	Weather and Climate Data	CSV, API	Global dataset (80-year history)	here API	High	Open-source, global coverage
Misinformation in Disaster Posts	GitHub (Mori Esam)	Misinfo social media data	CSV	1600 tweets	here Dataset	Medium	Seems to include COVID data as well
Claim Fraud Detection Dataset	Gomask AI	Insurance claim fraud data	CSV	100 records	here Synthetic dataset	Low	Requires an email to access
Disasters on social media	Kaggle	Misinfo social media data	CSV	~10,000 tweets	here Dataset	Medium	Need a Kaggle account, it doesn't show the tweets, just the ID
OpenPhish	OpenPhish	Phishing threat intelligence	CSV, JSON, text	Continuous live feed	here Feeds/dataset	High	Limited academic access period
Spamhaus	Spamhaus	Cyber threat intelligence (IP, spam sources)	API, DNSBL	Large-scale threat data	here API/dataset	High	Widely used in cybersecurity industry
DirectConnect API	Open Threat Exchange	Cyber threat intelligence	CSV	Provides daily updates	here API	High	Requires account to acquire key, also provides an endpoint security service
Malware Information Sharing Platform (MISP)	Malware Information Sharing Platform	Cyber threat intelligence	JSON, XML	Provides continuous updates	here Feeds here API	High	MISP provides both an API and OSINT RSS Feeds

4.3 Key Findings

Which datasets are usable immediately?

- OpenPhish and Spamhaus: These offer organized and frequently updated cyber threat intelligence data that can be used right away to identify malicious activity and phishing attacks.
- Misinformation in Disaster Posts (GitHub): This labelled dataset can be used directly to train classification models to identify misinformation during disasters.
- Weather API, BoM RSS feeds, and Open-Meteo: These provide dependable, regularly updated real-time weather and disaster data that is simple to integrate for hazard condition monitoring.
- NASA FIRMS: This is useful for monitoring and analyzing bushfires since it offers almost real-time fire detection data along with location and time information.
- CFA RSS Feeds: provides quick updates for CFA related incidents for practically real-time fire detection, providing incident information such as fire status, crews responding, and more. Can be used in conjunction with NASA FIRMS and BoS for detailed bushfire information.
- DirectConnect API: useful for collecting threat intelligence information from OTX, though this API does seem more suited for detecting threats targeting the service you are creating.
- MISP: Since they provide both feeds and APIs it can be better

- Which datasets have limitations?

- Existing datasets cover either disaster data or cyber threats, but not both together. This makes it difficult to train models for detecting cyber threats during disaster events.
- Social media disasters (Kaggle): This might only offer tweet IDs rather than actual content, necessitating extra steps to retrieve and process the data.
- NASA FIRMS: Its use for more in-depth analysis is limited by its lack of comprehensive contextual data, such as damage levels, casualties, or complete incident reports.
- Claim Fraud Detection Dataset: This is a small, artificial dataset that might not accurately reflect fraud or scam activity in the real world.
- EM-DAT: This dataset is less appropriate for real-time detection systems since it is historical and does not offer real-time updates.
- Both CFA RSS feeds and VicEmergency: Feeds only included data for Victoria and the southern part of New South Wales **(NOTE: emergency services RSS feeds can be acquired for every state and territory in Australia)**

- Any gaps in data availability?

- Disaster and cyber threat data cannot be combined into a single dataset, necessitating the integration of several datasets from various sources.
- Labelled datasets that concentrate on cyberthreats related to disasters, like phishing or scams that take place during disaster events, are scarce.
- Effective dataset combining for analysis is hampered by variations in data formats, update frequency, and structure.

4.4 Recommended Datasets

Weather API, Bureau of Meteorology (BoM) RSS feeds, CFA RSS feeds, NASA FIRMS, OpenPhish, Spamhaus, and the Misinformation in Disaster Posts datasets are the **best datasets** for the PHOENIX project, according to an analysis of usability, constraints, and data challenges. To accomplish the project's goal of combining hazard data with digital risk detection, these datasets offer a potent combination of real-time disaster data and cyber threat intelligence.

Real-time monitoring of disaster conditions is possible with the help of Weather API and BoM, which provide dependable and regularly updated environmental data, . This is reinforced by location-based bushfire detection data provided by NASA FIRMS and known incident details, status, and news provided by CFA RSS.

On the cyber front, OpenPhish and Spamhaus offer top-notch, regularly updated threat intelligence data that makes it possible to identify malicious activity and phishing attacks. Furthermore, the Misinformation in Disaster Posts dataset is useful for employing classification models to detect false information during disaster events. These datasets are the best options for supporting anomaly detection and cyber risk prediction in the PHOENIX framework because they are useful, pertinent, and appropriate for integration.

4.5 Data Challenges

1. Weather Data Services (BoM)

- Because the dataset is supplied in a variety of formats, including RSS, XML, and HTML, extra preprocessing is needed to standardize and incorporate the data into a single system.
- It offers real-time updates, but it can be technically difficult to manage continuous data feeds and guarantee consistent parsing.

2. NASA FIRMS

- The dataset mostly offers information on the location and intensity of fires, but it is devoid of specific details like damage or impacted populations.
- Utilizing APIs is necessary to access extended historical data, and time and location alignment is necessary to integrate it with other datasets.

3. EM-DAT

- The use of this dataset for real-time monitoring systems is limited because it is historical and does not offer real-time updates.
- Integration with live datasets may become more complicated due to variations in structure and time format.

4. Weather API (WeatherAPI.com)

- The dataset depends on API access, which, depending on usage, may have restrictions and request limits.
- It generates massive amounts of time-series data that need to be processed and stored effectively.

5. CFA RSS

- These feeds only offer data on CFA related incidents, and multiple feeds are required for all regions of Australia.
- These RSS feeds update constantly and inconsistently, where some feeds may update while others may not, requiring constant updates for RSS feeds.

6. VicEmergency

- Accessing the data collected from this service requires express permission being granted by them, making it less than feasible for immediate integration and use.
- This service only operates in Victoria and the southern parts of New South Wales, which could lead to data imbalance if implemented.

7. Open-Meteo

- The project requires filtering and choosing pertinent regional data because the dataset contains global data.
- When merging with other datasets, disparities in data resolution and format could lead to inconsistencies.

8. Misinformation in Disaster Posts (GitHub)

- Due to its small size, the dataset might not accurately reflect patterns of misinformation in the real world.
- It needs to be cleaned before use as it might contain inconsistent or unnecessary data.

9. Disasters on Social Media (Kaggle)

- Certain dataset versions only offer tweet IDs; additional API calls are needed to obtain the actual content.
- Preprocessing and text cleaning are necessary for analysis because the data is noisy and unstructured.

10. Claim Fraud Detection Dataset

- Since this dataset is artificial, it might not fairly represent fraudulent activity in the real world.
- Because of its small size, the dataset is not very useful for training reliable machine learning models.

11. OpenPhish

- The dataset excludes more comprehensive contextual information about attacks and only concentrates on phishing URLs.
- Managing streaming data and regular data refreshes are necessary for continuous updates.

12. Spamhaus

- The dataset's use of specialized formats, like DNS-based lists, complicates integration.
- It primarily concentrates on threats at the network level, which might not be directly related to cyber behavior at the user level.

13. DirectConnect API

- Threat intelligence comes from a wide range of different individuals submitting information to OTX, leading to varying levels of trustworthiness.
- This API is an American product, so most information provided will be for Americans and may or may not be useful for PHOENIX’s goals

14. Malware Information Sharing Platform (MISP)

- MISP may require many different tools to configure it properly, such as MISP Puppet module or MISP-Sizer

5. Model Evaluation

5.1 Model Overview

Classification:

This machine learning technique is utilized to predict labels and categories based on a series of input data. As suggested by GeeksForGeeks (2016) the goal of this model is to assign input data into predefined classes, such as distinguishing between legitimate and fraudulent communications within the Phoenix system. In the context of the Phoenix project and machine learning, classification can be used to learn specific data patterns, such as sudden increases in malicious data entries during fires / storms, and associate classes accordingly. Classification utilizes common algorithms such as Logistic Regression, Random Forest, and SVM. These classification models can have **high accuracy and clear outputs** (class labels). However, classification models rely on labelled data and may struggle to detect new or previously unseen attack patterns.

Anomaly Detection:

Anomaly Detection models specialize in detecting abnormal data types and patterns that deviate from normal behaviour. As described through Anomaly Detection in Machine Learning, Calsoft Inc (2026), anomalies are data points that significantly deviate from normal behaviour and are flagged by specialised algorithms. For example, these models can detect spikes in CPU servers, longer transactions, and abnormal entry locations. E.g. A user may submit an entry from Melbourne at 10:00pm, and then from Paris at 10:05pm. These models can learn a safe baseline from data history, improving pattern recognition over time. **Strengths** include protection from zero-day attacks, and labels are not necessary. A key limitation is a higher likelihood of false positives, as unusual but legitimate behaviour may also be flagged.

Time-Series:

This model specializes in analyzing data over time, capturing different trends and patterns to improve forecasting capabilities. While this model type is typically associated with forecasting models such as Long Short-Term Memory and the Autoregressive (AR) Model, time-series models can also be integrated with anomaly detection and machine learning (Aarthy R. (2024). Time-series models differentiate from the other models though, as they don't directly depend on independent and unrelated data points, such as classification models. With this model, "time is the only independent variable" (Belcic, I., & Stryker, C (2025). Ultimately, time-series models focus on **when** things will happen instead of just what events will occur, which could include forecasts of a gradual increase in spam emails over the next 2 years, or seasonal fraud patterns over summer. **Strengths** include high levels of detection in pattern changes over longer periods of time, and time-series models also excel in capturing trends / seasonality. The model is **limited** by high complexity, the need for high-level time-based data. However, they are less effective at detecting sudden or unexpected anomalies.

Probabilistic Risk:

Supported by NASA (Boyer, R. (2018)), probabilistic risk assessment (PRA) incorporates both qualitative and quantitative views to provide structured evaluation of risks within systems and datasets. These models specialize in estimating the probability of certain events, rather than giving simple yes / no outputs. As a key example in relation to the Phoenix project, a probabilistic risk model such as Bayesian Networks / Logistic Regression could output an 84% chance that spam email is fraudulent or malicious. PRA models can also enhance risk management and decision making by providing a sensitive and probability-based analysis of different options and datasets. (Hetes, B. et al, n.d). **Strengths** include flexibility in decision-making, high applicability to real-world systems, and supportive of threshold tuning (e.g. flag if risk score > 0.8). **Limitations** include the need for calibration and regular testing, and that PRA can also be challenging to implement due to model complexity and required domain knowledge.

5.2 Model Comparison Table

Model Type	Use Case	Pros	Cons	Data Requirements	Suitability
Logistic Regression	Text Classification (Scam vs legitimate)	Simple, fast, highly interpretable, works well with text features.	Cannot capture complex non-linear relationships	Labelled text data	High
Random Forest	Scam/phishing detection	High accuracy, handles non-linear patterns, robust to noise.	Less interpretable, can be computationally heavier.	Structured labelled dataset	High
Isolation Forest	Anomaly detection	Detects unknown threats, does not require labelled data	May produce false positive, less intuitive results	Unlabelled or semi-labelled data	High
LSTM (Long Short-Term Memory)	Time-Series prediction	Captures temporal patterns, useful for sequential data	Complex, requires large dataset and training time	Time-series data	Medium
Bayesian Network	Probabilistic Risk	Handles uncertainty, provides probabilistic outputs	Complex to design, requires prior assumptions	Probabilistic structured data	Medium

5.3 Recommended Model(s)

Baseline Model: Random Forest (Classification)

For the PHOENIX system, Random Forest is chosen as the main baseline model. Binary classification problems, including differentiating between authentic and fraudulent messages, are a good fit for it. The model works well with structured and text-derived data, is resilient to noise, and can handle non-linear correlations. It is also appropriate for first implementation because it offers comparatively high accuracy without requiring a lot of adjustment.

Secondary Model: Isolation Forest (Anomaly Detection)

It is advised to use Isolation Forest as a supplementary model to identify unusual or unidentified cyberthreats. It is efficient at spotting odd patterns in cyber activity during crisis situations because, in contrast to supervised models, it does not significantly rely on labelled data. This is especially helpful when new kinds of assaults or frauds appear that weren't in the training sample. The Isolation model is a model that is a part of the PyOD Python module and can be used from there.

Future/Advanced Models:

Time-Series Model, or LSTM

In order to analyse temporal trends in cyber threats during disaster occurrences, Long Short-Term Memory (LSTM) networks are suggested for future development. Cyber hazards can be predicted as a crisis develops thanks to these models' ability to capture sequential dependencies and trends over time.

Bayesian Network (Modelling Probabilistic Risk)

An advanced method for probabilistic risk assessment is proposed: Bayesian Networks. The system can assess cyber risk levels based on a variety of parameters, including communication patterns, threat indicators, and the severity of disasters, thanks to these models' ability to account for uncertainty and interdependence between variables.

5.4 Justification

The models that are picked are determined by how well they fit the project's goals, the features of the dataset, and the viability of their implementation.

Practical Feasibility:
The baseline and secondary models (Random Forest and Isolation Forest) are appropriate for a capstone project with limited time and resources because they are reasonably easy to develop using common machine learning packages.

Alignment with Available Data:
While Isolation Forest can function on unlabelled or partially labelled data, addressing the limits of available datasets, Random Forest is best suited for tagged datasets, such as phishing or scam data.

Complementary Capabilities:
A more complete detection system is produced by combining anomaly detection with categorization. While anomaly detection finds novel or uncommon patterns, classification models deal with recognized dangers.

Extensibility and Scalability:
Future development of LSTM and Bayesian models will guarantee that the system may be expanded to accommodate probabilistic risk scoring and time-based forecasting, improving its long-term capabilities.

Balanced Trade-off Between Performance and Complexity:
The selected models ensure that the system maintains efficiency while producing dependable results by providing a good balance between accuracy, interpretability, and computational complexity.

6. Pre-Trained / Open-Source Models

Model Name	Use Case	Pros	Cons	Datatype	Computational Requirement	Suitability / Integration
Nvidia – cosmos-reason2-8b	Generates responses to physical video / imagery. Could be used to analyse forecasts.	Provides fast and accurate notes on real-world photos & video. Free to download and use.	Needs physical imagery / video. Requires correct prompts depending on the file type.	Works with images and videos. Detects both 2D and 3D "point localization."	Medium – high.	Low suitability. Project prioritizes datasets over imagery / video. Could be paired with other models.
Nvidia - corrdiff	Prediction model used to forecast data to produce and predict visual weather fields.	Ready for commercial use. Weather prediction.	Can mispredict fine-resolution features.	No data needed to operate. Uses AI to create forecasts.	Medium.	Low suitability. Perfect for forecasts. Could be paired with other models such as PyOD to detect anomalies.
AnomalyBERT	Time-based risk spikes	Self-supervised Strong at complex correlations	Harder explainability	Multivariate time-series numerical data.	High.	High Suitability Matches perfectly with ADCRS in project scope.
PyOD	Time-series anomaly detection	Includes 50 anomaly detection algorithms from simple to advanced Easy to install Python module	Potential issues on large datasets Varying quality in model performance Steep learning curve	multivariate time-series data.	Low-Medium. It depends on the algorithm used.	High Suitability Provides a wide range of likely suitable models for ADCRS implementation
DistilBERT	Text/Sentiment classification	A Distilled version of the BERT model, allowing for faster run times and lighter computation	Struggles with complex language tasks,	Numerical data	Low	Medium fine-tuned models do exist but are uncertain quality
HuggingFace: fraud-detection-models	Fraud-detection system. 11 ML models.	High accuracy, 6 models > 90%. E.g. Phishing (87.3%).	Requires other imports and minimal fine-tuning. Dataset will need to be imported.	Data types may vary. Numerical & categorical.	Medium	Medium-High suitability. May require fine-tuning but has high accuracy for anomaly detection.
HuggingFace: fake-news-detector	Fake news detection	High accuracy model fine-tuned from Roberta-large	Focuses on news articles and might not be tuned to misinformation from comments/tweets	Numerical/Boolean data	Medium	Medium This model is only designed as a screening tool for fake news, only classifying it as either real or fake.

cardiffnlp/twitter-roberta-base	Social media / disaster misinformation detection	Trained on Twitter data Strong contextual understanding	Requires fine-tuning for domain-specific tasks	Text	Medium	Perfect for disaster-related messages and public communication
sentence-transformers/all-MiniLM-L6-v2	Message similarity / scam pattern detection	Fast and lightweight Detects duplicate or similar scam messages	Not a direct classification model	Text embeddings	Low	Useful for clustering repeated scam patterns

Locations:

Nvidia – Cosmos-reason2-8b: https://build.nvidia.com/nvidia/cosmos-reason2-8b

Nvidia – Corrdiff: https://build.nvidia.com/nvidia/corrdiff

AnomalyBERT: ondrejspilka/anomalyBert · Hugging Face

PyOD: https://pyod.readthedocs.io/en/latest/

DistilBERT: https://huggingface.co/distilbert scam fine-tuned: here here misinfo fine-tuned: here here

HuggingFace (fraud-detection-models): https://huggingface.co/vaibhav07112004/fraud-detection-models

HuggingFace (fraud-news-detector): https://huggingface.co/ThomasTschinkel/fake-news-detector

cardiffnlp/twitter-roberta-base: https://huggingface.co/cardiffnlp/twitter-roberta-base?

sentence-transformers/all-MiniLM-L6-v2: https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

6.1 Performance & Scalability Evaluation

Nvidia – Cosmos-reason2-8b: This model has strong multi-media capabilities, working with images, videos and texts. Has a high level of visual understanding, although the performance heavily depends on the input quality and prompt engineering. It appears that the model needs specific prompts and imported datasets as well, which increases the amount of training and fine-tuning from the team. For the Phoenix project, cosmos-reason2-8b has poor scalability due to the GPU-heavy infrastructure. The data pipelines would need to be tuned for media processing, which would require additional preprocessing layers to reduce computational stress.

Nvidia – corrdiff: This forecasting model is designed for high-resolution weather prediction, which excels at macro-level forecasts. In relation to fraud / risk detection, this model would be weak in identifying the fine-grained anomalies. This framework has moderate scalability, with efficient runtimes once deployed. However, this model would not be flexible for anomaly detection pipelines, and would require specialized climate-related datasets, which is not a focus of the Phoenix project.

AnomalyBERT: Prioritizes temporal dependencies, with complex correlations between variables. This model is also self-supervised, reducing the need for labelled data. AnomalyBERT also has strong levels of detection for subtle anomalies, ideal for ADCRS. Higher levels of scalability, with less dependence on computer infrastructure. This model works well with large sets of time-series data; however the model may need fine tuning (includes training costs).

PyOD: Highly flexible *library* with models like isolation forest and random forest available for download. This collection of models varies on flexibility, with some requiring more / less resources. Some models can be lightweight and more applicable for the Phoenix project; however the accuracy of their detection is often lower. Due to all the different factors (including the need for fine-tuning), the PyOD library has mixed levels of scalability. Simple projects with low-levels of tuning could be scaled well in future, however complex models could struggle when scaling to larger datasets.

DistilBERT: DistilBERT provides good performance for text classification tasks while maintaining low computational requirements. It is suitable for processing large volumes of message data within the integrated dataset, enabling efficient classification of scam and legitimate communications. Its lightweight architecture allows faster inference, making it appropriate for near real-time systems.

However, its performance may be lower compared to larger transformer models when handling complex or ambiguous language. Additional fine-tuning may be required to improve accuracy for disaster-specific scam detection.

HuggingFace – Fraud Detection Models: This collection of models demonstrates strong performance across multiple fraud detection tasks, with several models achieving high accuracy levels. It is suitable for identifying structured fraud patterns and anomaly-based behaviour within the integrated dataset.

However, these models may require additional preprocessing and dataset alignment to ensure compatibility with the PHOENIX pipeline. Fine-tuning may also be necessary to adapt the models to disaster-specific cyber threats.

HuggingFace – Fake News Detector: This model is effective for identifying misinformation and fake content, making it relevant for detecting misleading messages during disaster events. It provides strong classification performance when applied to structured news-like text data.

However, its performance may be limited when applied to informal or short-text formats such as social media posts or messages. Additional fine-tuning or preprocessing may be required to adapt it to real-time disaster communication scenarios

cardiffnlp/twitter-roberta-base : This model performs well when analyzing text from social media, especially when brief messages and colloquial language are included. It is quite good at spotting trends in communications about disasters, including false information and scam messages. Scalability-wise, the model works well with big text datasets, but inference and fine-tuning demand a moderate amount of computing power. It can be integrated into the PHOENIX system because of its capacity to manage large amounts of real-time data, particularly when analyzing public communication during crisis occurrences.

sentence-transformers/all-MiniLM-L6-v2: This model performs well in tasks involving embedding generation and text similarity. It can analyze big datasets fast with little processing power because it is scalable and lightweight. In the merged dataset, the model works very well at identifying recurring or comparable scam messages. While it does not perform direct classification, its ability to support clustering and similarity-based analysis makes it a valuable addition to the overall system. It is very well suited for scalable deployment due to its minimal resource needs.

6.2 Feasibility Checks

Data compatibility, implementation complexity, and computational requirements within the PHOENIX project were used to assess the viability of the chosen models.

It is very possible to handle text and structured data from the integrated dataset using models like DistilBERT, Twitter-roberta-base, and Random Forest. These models can be effectively applied for classification tasks like identifying fraudulent and authentic messages, are backed by widely accessible libraries, and demand little processing resources.

Similarly, because of their versatility in managing anomaly detection and similarity-based tasks, PyOD, Isolation Forest, and all-MiniLM-L6-v2 are also practical. These models are useful for spotting odd trends and recurring fraudulent behaviors in the dataset and can function with partially labeled or unlabeled data.

Strong performance for time-series and anomaly detection makes models like AnomalyBERT somewhat practical, but they need more processing power and fine-tuning. When the system integrates temporal data, these models are better suited for later phases of development.

In contrast, models such as Nvidia Cosmos and Corrdiff have low feasibility, as they rely on image, video, or forecast-based data that does not align with the integrated dataset, which primarily consists of text and structured cyber threat information.

Overall, the selected models are practical and achievable within project scope, with a strong balance between performance, scalability, and implementation feasibility.

6.3 Model Readiness and Implementation Summary:

Model / Framework	Primary Function	Implementation Source	Complexity
DistilBERT	Text Classification	Hugging Face/ Transformers	Low
Twitter-roberta-base	Sentiment and Scam analysis	CardiffNLP / GitHub	Low
Isolation Forest	Anomaly Detection	Scikit-learn /PyOD	Low
All-MiniLM-L6-v2	Semantic Similarity	Sentence=Transformers	Medium
Random Forest	Structured Fraud Detection	Scikit-learn	Low
AnomalyBERT	Time-Series Risk	Research Repo	high
NVIDIA NIM APIs	Scalable Inference	Build.nvidia.com	Medium

6.4 Mapping Models to Pipeline

The PHOENIX system processes the combined dataset using a layered pipeline in which several models are applied at various stages. First, information about cyber threats and disasters is gathered from many sources and merged into a single dataset. After that, preprocessing and feature engineering are used to clean and prepare this dataset.

Models like DistilBERT, cardiffnlp/twitter-roberta-base, and the fake-news detector can be used to analyze message content at the text analysis stage and find patterns connected to scams, disinformation, or suspicious language. The fraud-detection models and Random Forest can be used to categorize records as suspicious or authentic for structured classification tasks.

While sentence-transformers/all-MiniLM-L6-v2 can help similarity checking by identifying recurring scam patterns or closely comparable messages, Isolation Forest and/or other modules from PyOD can be employed at the anomaly detection stage to discover unusual or previously undetected cyber threat behavior.

In the future, models like Corrdiff might support catastrophe forecasting context instead of direct threat detection, while AnomalyBERT might be used to analyze time-based cyber risk spikes. Alerts, classifications, and risk insights that can aid in PHOENIX system decision-making are the pipeline's final outputs.

7. Overall Recommended Approach

Developing an integrated machine learning pipeline with a single dataset that integrates cyber threat intelligence and disaster context data is the suggested strategy for the PHOENIX system. Instead of analyzing cyber and disaster data independently, this method enables the system to identify cyberthreats within disaster scenarios.

The system's first implementation should concentrate on a workable hybrid strategy that combines anomaly detection and text classification. While Random Forest can be used as a baseline classification model to distinguish between authentic and fraudulent communications, text-based models like **DistilBERT** can be used to analyze message content. Then, as secondary models, Isolation Forest or other PyOD algorithms can be used to identify uncommon or hitherto undiscovered cyberthreats.

This approach is recommended because it aligns with the available data, is feasible within the project scope, and provides scalability for future improvements. Advanced models such as **AnomalyBERT**, **LSTM**, and **Bayesian Networks** can be explored in later stages for time-based analysis and probabilistic risk scoring.

8. Integration with Other Tasks

- AI002 outputs (selected datasets and recommended models) will directly support **AI004 (Feature Engineering)** by defining which features should be extracted, such as text patterns, URL indicators, and disaster related variables.
- The selected models (Random Forest, DistilBERT, AnomalyBERT) will be implemented in **AI007 / AI010 (Model Development**), ensuring consistency between research and development phases.
- Identified data gaps will inform **AI005 (Synthetic Data Generation)**, where artificial datasets can be created to simulate disaster-related cyber threats.
- Integration with backend systems is required to ensure real-time data ingestion from **APIs (e.g., BoM, OpenPhish)** and proper data formatting.
- A unified schema must be developed collaboratively with backend teams to ensure **compatibility, scalability, and efficient data flow across the Phoenix system.**

9. Conclusion

To aid in the development of the PHOENIX system, this research examined pertinent datasets and machine learning models. It found that although trustworthy sources of cyber threat intelligence and catastrophe data are accessible, there is a large lack of linked datasets that merge both fields.

According to the review, a viable and efficient method for identifying cyberthreats during catastrophic situations is a classification-based approach utilizing Random Forest, bolstered by anomaly detection with Isolation Forest. These models offer a compromise between performance, interpretability, and implementation practicality and are in good agreement with the data that is already accessible.

A hybrid dataset method, which combines several data sources and incorporates synthetic data generation to mimic realistic disaster-related cyber threat scenarios, is suggested as a solution to the current data restrictions. This strategy guarantees that the system will continue to be flexible and scalable for upcoming improvements.

Overall, the PHOENIX project demonstrates strong technical feasibility and provides a scalable foundation for an AI-driven cybersecurity system. By integrating disaster intelligence with cyber threat detection, the system has the potential to significantly improve real-time risk awareness and response during critical events.

10. References

Include dataset links, APIs, and research papers. APA7 Referencing Style.

Aarthy R. (2024, May 21). Time Series Analysis: Forecasting Models and Anomaly Detection. Medium. https://aarthy-botcampusai.medium.com/time-series-analysis-forecasting-models-and-anomaly-detection-f72a6adb123f

Anomaly Detection in Machine Learning: Classification Algorithms vs Anomaly Detection | Calsoft Inc. (2026). Calsoftinc.com. https://www.calsoftinc.com/blog/machine-learning-classification-algorithms-vs-anomaly-detection

Belcic, I. (2024, October 15). Classification in Machine Learning. Ibm.com. https://www.ibm.com/think/topics/classification-machine-learning

Belcic, I., & Stryker, C. (2025, June 5). Time Series Model. Ibm.com. https://www.ibm.com/think/topics/time-series-model

Blocklists | Trusted, actionable DNSBLs for email filtering. (2026). The Spamhaus Project. https://www.spamhaus.org/blocklists/

Boyer, R. (2018). Probabilistic Risk Assessment (PRA): Analytical Process for Recognizing Design and Operational Risks Prepared by Chief, Risk & Reliability Analysis Branch NASA Johnson Space Center Safety & Mission Assurance (S&MA) Prepared for Equinor Oil & Gas. https://ntrs.nasa.gov/api/citations/20200001598/downloads/20200001598.pdf

Cardiff NLP. (n.d.). Twitter RoBERTa base. Hugging Face https://huggingface.co/cardiffnlp/twitter-roberta-base

Free Open-Source Weather API | Open-Meteo.com. (n.d.). Open-Meteo.com. https://open-meteo.com/

GeeksforGeeks. (2016, June 15). Getting started with Classification. GeeksforGeeks. https://www.geeksforgeeks.org/machine-learning/getting-started-with-classification/

GoMask.ai. (2025, July 12). Claim Fraud Detection Dataset. GoMask.ai. https://gomask.ai/marketplace/datasets/claim-fraud-detection-dataset

Hetes, B., Olsen, M., Gallagher, K., Stahl, C., & Schoeny, R. (n.d.). *Using Probabilistic Methods to Enhance the Role of Risk Analysis in Decision Making Managers' Summary*. Retrieved March 26, 2026, from https://19january2017snapshot.epa.gov/sites/production/files/2014-11/documents/pra_mgr_summary_0609.pdf

Klaas, J. (2018). Disasters on social media. Kaggle.com. https://www.kaggle.com/datasets/jannesklaas/disasters-on-social-media

moriesam. (2025). GitHub - moriesam/Misinformation-in-Disaster-Related-Social-Media-Posts-: Classification model(s) that help Emergency Management Agencies identify incorrect information before, during and after disasters. GitHub. https://github.com/moriesam/Misinformation-in-Disaster-Related-Social-Media-Posts-?tab=readme-ov-file

NASA-FIRMS. (n.d.). Localhost.modaps.eosdis.nasa.gov. https://firms.modaps.eosdis.nasa.gov/active_fire/

(2020). Emdat.be. https://public.emdat.be/

NVIDIA. (n.d.). Cosmos Reason2 8B. https://build.nvidia.com/nvidia/cosmos-reason2-8b

NVIDIA. (n.d.). CorrDiff. https://build.nvidia.com/nvidia/corrdiff

Reimers, N., & Gurevych, I. (n.d.). All-MiniLM-L6-v2. Hugging Face https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (n.d.). DistilBERT. Hugging Face https://huggingface.co/distilbert

Spilka, O. (n.d.). AnomalyBERT. Hugging Face https://huggingface.co/ondrejspilka/anomalyBert

Tschinkel, T. (n.d.). Fake news detector. Hugging Face https://huggingface.co/ThomasTschinkel/fake-news-detector

Vaibhav07112004. (n.d.). Fraud detection models. Hugging Face https://huggingface.co/vaibhav07112004/fraud-detection-models

Weather API. (2021). Free Weather API - WeatherAPI.com. Weatherapi.com. https://www.weatherapi.com/

Weather Data Services. (2025). Bom.gov.au. https://reg.bom.gov.au/catalogue/data-feeds.shtml

Zhao, Y. (n.d.). PyOD: Python toolbox for scalable outlier detection. https://pyod.readthedocs.io/en/latest/